PhytoTarget

Bias-corrected target prediction for plant natural products, with source-organism enrichment.

Traditional medicine names plants. Drug discovery needs molecules and targets.

This closes that gap in the direction a field researcher can act on.

ChEMBL (CC BY-SA 3.0) + LOTUS/Wikidata (CC0) + RDKit | research prototype, not clinical advice

THE GAP

Centuries of observation, indexed by the wrong key

Traditional knowledge is organism-level. It records that a preparation of a plant helps a condition. It does not name the molecule or the protein it acts on.

Drug discovery is molecule-level. Target prediction, assays and medicinal chemistry all operate on structures. The two bodies of knowledge do not share an index.

The bridge exists but is unexploited. LOTUS, via Wikidata, holds hundreds of thousands of referenced plant-compound occurrence statements. Joined to a target model, it turns a plant into a set of testable molecular hypotheses.

And the inverse question is the one nobody tools for:

“Which plants are over-represented among the predicted actives?”

That answer is a collection plan, not just a compound list.

THE REAL PROBLEM

A screening model is easy to make look excellent

Four failure modes produce impressive numbers that mean nothing. This pipeline instruments all four instead of assuming they are absent.

FAILURE MODE	WHAT IT DOES TO YOUR METRICS	WHAT WE DO
Random train/test split	Analogues of one scaffold land in both halves; the model is graded on chemistry it memorised.	Scaffold split is the headline. Random is reported beside it, so the inflation is visible.
Hidden label leakage	Any bug that lets labels reach features inflates everything downstream.	y-scrambling: 10 refits on permuted labels must collapse to 0.50.
Artificial decoy geometry	Actives separable from inactives by nearest-neighbour distance alone.	AVE bias (Wallach & Heifets 2018) computed and reported per target.
Publication-biased prevalence	ChEMBL is 50-70% actives, which caps EF@1% near 1.6 - a saturated EF looks like a triumph.	Held-out actives hidden in the real 23k natural-product library instead.

ARCHITECTURE

One standardisation path, two sources, four guards



EVERY RANKED HIT CARRIES

Applicability domain max Tanimoto to training; below threshold is flagged, not silently ranked

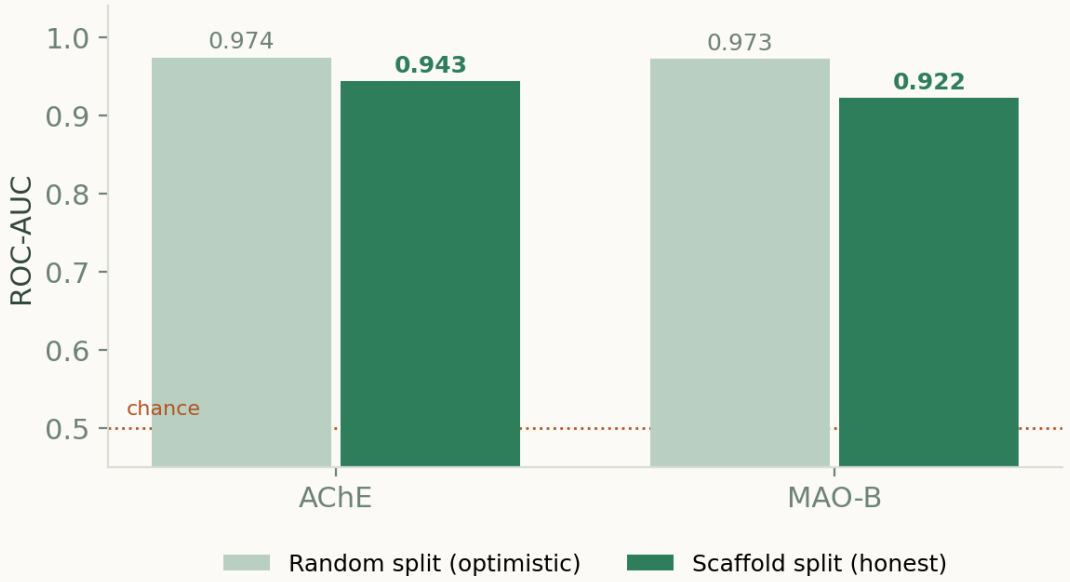
PAINS count frequent-hitter substructures, so a promiscuous polyphenol is not celebrated

Known-active flag rediscoveries are kept as validation but excluded before enrichment testing

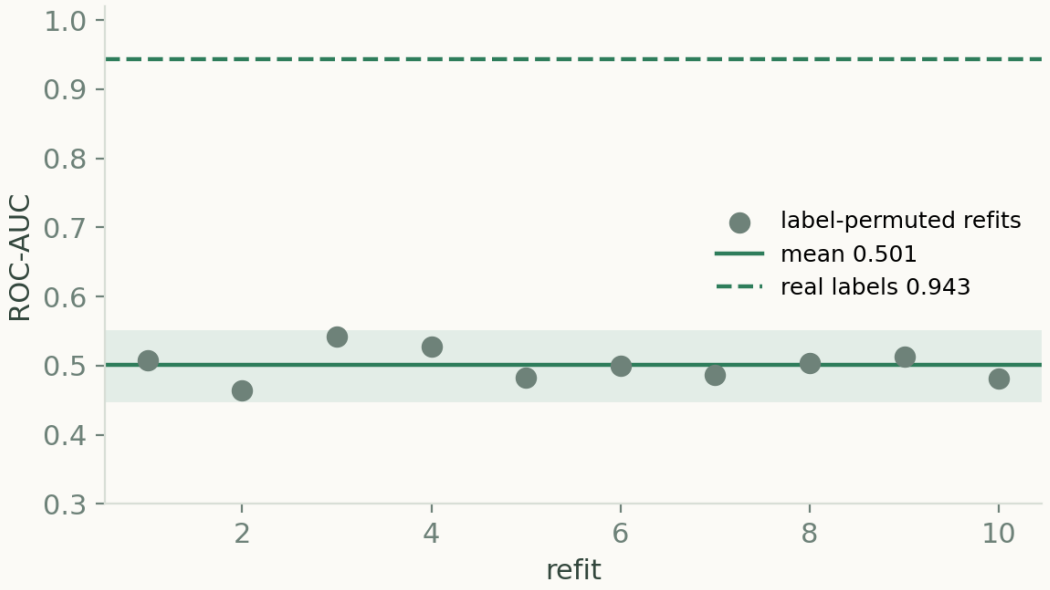
VALIDATION

The honest number, next to the flattering one

Scaffold split vs random split



y-scrambling: the leakage control



Random splitting overstates AChE performance by +0.030 AUC and MAO-B by +0.050. With labels permuted, performance collapses to 0.501 ± 0.022 over 10 refits - tight on chance, so there is no leakage path in the pipeline. That control is what makes the 0.943 worth reading.

WHAT WE ARE NOT HIDING

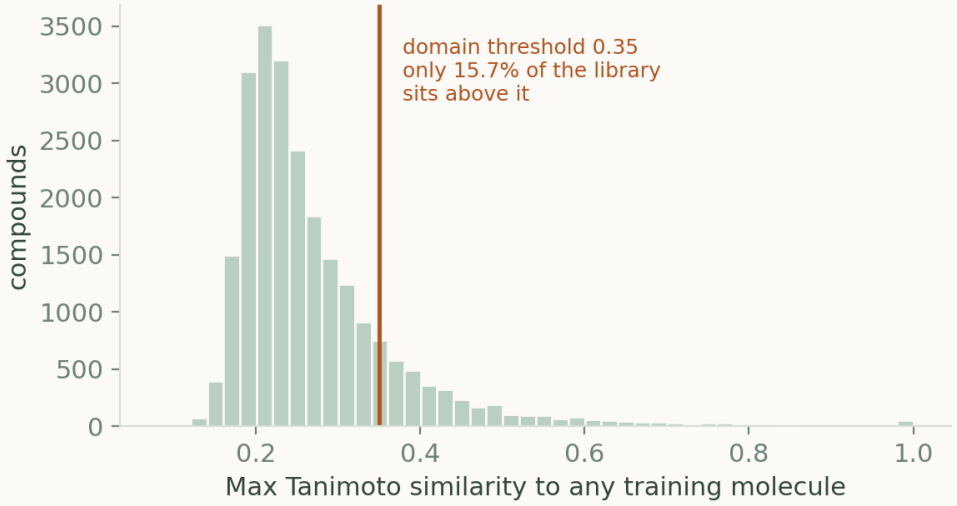
Two results that argue against us

1. AVE split bias is +0.50, which is middling, not good.

A meaningful part of the scaffold split is separable by nearest-neighbour geometry alone. We report it rather than re-splitting until the number improves: tuning a split to minimise AVE and then quoting the resulting AUC is itself a way of overfitting to a validation statistic.

2. Only 15.7% of the plant library is inside the applicability domain.

Natural-product scaffolds are mostly unlike the synthetic chemistry in ChEMBL. This is the single most important caveat in the whole project, and it is a property of the problem rather than a bug in the model. The honest response is to sur compounds only - not to quietly rank 23,761 extrapolations.



OUTPUT

A ranked list that tells you how much to trust it

23,765
source-organism compounds scored

3,726
novel + inside applicability domain

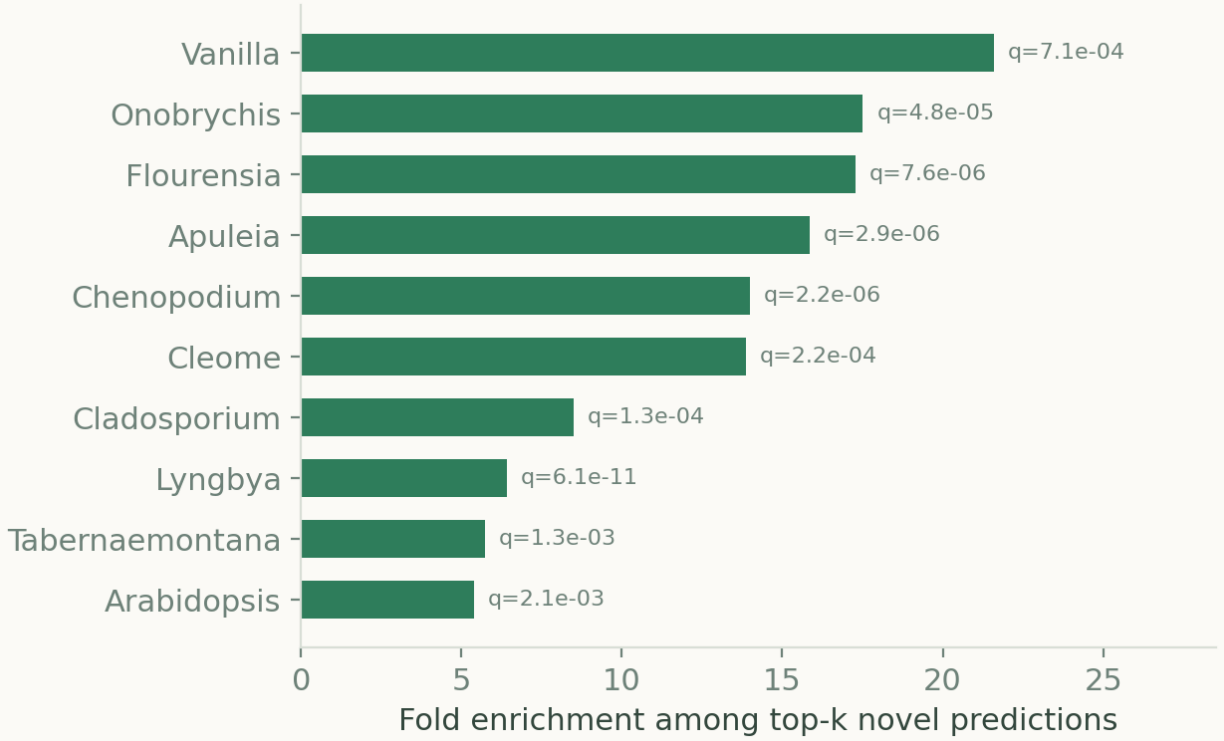
3,265
of those with no PAINS alert

8
known actives recovered (flagged, excluded from enrichment)

#	COMPOUND	P(ACTIVE)	TRAIN SIM.	PAINS	SOURCE PLANTS
5	**terreulactone D**	**0.843**	0.74	0	*Aspergillus terreus*
7	**Territrem E**	**0.818**	0.87	0	*Aspergillus terreus*
8	**(2S)-2,6-dimethyl-1-oxo-2,3-dihydroinden…**	**0.809**	0.46	0	*Croton steenkampianus*
15	**methyl (1R,12S,14R,15S)-15-ethyl-12-[(1R…**	**0.777**	0.82	0	*Tabernaemontana bovina*
19	**Daphnoretin methyl ether**	**0.743**	0.39	0	*Ruta graveolens, Ruta montana*
24	**herbertene**	**0.736**	0.39	0	*Herbertus aduncus, Herbertus dicranus, Herbertus nor…*
33	**steganolide A**	**0.712**	0.37	0	*Steganotaenia araliacea*
36	**Isoterreulactone A**	**0.711**	0.80	0	*Aspergillus terreus*
38	**Methyl galbanate**	**0.699**	0.36	0	*Ferula assa-foetida, Ferula sinkiangensis*

THE INVERSE QUESTION

Which plants are over-represented among the hits?



Hypergeometric over-representation of source taxa in the top-k predictions, Benjamini-Hochberg corrected across every taxon tested - the same test used for GO-term enrichment.

Known actives are removed before testing. Otherwise a genus scores as “enter enriched” purely because its alkaloid is already a licensed drug the model memorised, which tells a prospector nothing new.

62

taxa significant at FDR < 0.05, of 12,203 tested

Taxa with too few library compounds are not tested at all: the test has no power there and including them only inflates the multiple-testing burden.

WHY, NOT JUST WHAT

Every edge carries its evidence

A ranked CSV answers “what should I test”. It does not answer “why”, and “why” is what makes a result arguable in front of a pharmacognosist.



CONTAINS a referenced LOTUS/Wikidata occurrence statement (P703), not an assertion.

PREDICTED_ACTIVE the calibrated probability and the applicability-domain flag together.

KNOWN_ACTIVE the measured pChEMBL value, where one exists.

IMPLICATED_IN a curated target-disease association.

So a path through the graph is a chain of individually checkable claims rather than an opaque score. Exports to GraphML for Cytoscape or Gephi.

LIMITATIONS

What this does not do

Ligand-based models interpolate. They predict from resemblance to known actives. Genuinely novel scaffolds fall outside the domain - exactly where interesting natural products sit. The flag makes that visible; it does not solve it.

Decoys are presumed inactive, not measured inactive. An unlabelled library compound that is truly active counts against the model, so that estimate is conservative.

A predicted binder is not a drug. Nothing here addresses absorption, metabolism, toxicity, tissue exposure, or whether engaging the target helps a patient.

Occurrence is not bioavailable dose. A compound reported from a plant says nothing about concentration, seasonal variation, or whether a traditional preparation extracts it at all.

Enrichment inherits study bias. Well-studied genera have more library compounds and more opportunity to appear enriched. The hypergeometric null accounts for how many, not for why.

Ethnobotanical notes are never model input. They are literature context for human interpretation, so they cannot leak into the statistics they sit beside.

Research prototype. Not clinical advice.

Traditional knowledge referenced here originates with the communities who developed and maintained it over generations; this tool points back toward that knowledge rather than claiming it. Data: ChEMBL (CC BY-SA 3.0), LOTUS via Wikidata (CC0), RDKit (BSD-3).